

RESEARCH ARTICLE

Responses of population structure and genomic diversity to climate change and fishing pressure in a pelagic fish

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Abstract

The responses of marine species to environmental changes and anthropogenic pressures (e.g., fishing) interact with ecological and evolutionary processes that are not well understood. Knowledge of changes in the distribution range and genetic diversity of species and their populations into the future is essential for the conservation and sustainable management of resources. Almaco jack (*Seriola rivoliana*) is a pelagic fish with high importance to fisheries and aquaculture in the Pacific Ocean. In this study, we assessed contemporary genomic diversity and structure in loci that are putatively under selection (outlier loci) and determined their potential functions. Using a combination of genotype–environment association, spatial distribution models, and demogenetic simulations, we modeled the effects of climate change (under three different RCP scenarios) and fishing pressure on the species' geographic distribution and genomic diversity and structure to 2050 and 2100. Our results show that most of the outlier loci identified were related to biological and metabolic processes that may be associated with temperature and salinity. The contemporary genomic structure showed three populations—two in the Eastern Pacific (Cabo San Lucas and Eastern Pacific) and one in the Central Pacific (Hawaii). Future projections suggest a loss of suitable habitat and potential range contractions for most scenarios, while fishing pressure decreased population connectivity. Our results suggest that future climate change scenarios and fishing pressure will affect the genomic structure and genotypic composition of *S. rivoliana* and lead to loss of genomic diversity in populations distributed in the eastern-central Pacific Ocean, which could have profound effects on fisheries that depend on this resource.

KEYWORDS

aquaculture, biomod2, CDMetaPop, fishery, future sea climate scenarios, genetic populations simulations, Pacific Ocean

1 | INTRODUCTION

Anthropogenic climate change drives several major changes in marine ecosystems, including increasing sea temperatures, changes in dissolved oxygen concentration, and decreases in ocean pH (Bahri et al., 2018; Mora, Wei, et al., 2013; Worm & Lotze, 2021). The contemporary and future impacts of these changes on marine species are not always well understood, but modeled simulations can be

used to better understand the impact of climatic changes on adaptation and genomic vulnerability (Cheung et al., 2009; Nielsen et al., 2021), and for predicting the spatial distributions of species under projected future conditions (Melo-Merino et al., 2020; Moss et al., 2010; Poloczanska et al., 2016).

In marine ecosystems, environmental factors such as salinity and temperature may act as selective pressures that can trigger behavioral, physiological, and molecular adjustments to cope with environmental

changes, for example, in teleost fishes (Dalongeville et al., 2018; Poloczanska et al., 2016; Selkoe et al., 2016). Species' responses to changing oceanic conditions are varied and include changes in the water depth range they inhabit, growth, reproduction, and mortality rates, as well as impacting productivity and trophic interactions and functional trait diversity (McLean et al., 2019; Pinsky et al., 2013; Poloczanska et al., 2013, 2016; Pörtner & Peck, 2010). The responses of species to environmental changes should also involve the understanding of evolutionary processes (Bernatchez, 2016). This approach allows for better understanding of changes in genetic diversity, genetic structure, migration patterns and the evolutionary potential of populations (Bernatchez, 2016; Davis et al., 2005; Gerber et al., 2014; Hodapp et al., 2023), that support evidence-based decision-making to manage marine resources (Nielsen et al., 2021, 2023). Changes in the distributional ranges of exploited species are particularly important in a management context, since they may negatively affect the biomass of the exploited resources and impact human livelihoods (Möllmann et al., 2021; Ovenden et al., 2015; Pecl et al., 2017; Pinsky & Palumbi, 2014). Future models predict that for many regions the maximum catch potential in exclusive economic zones will decrease by approximately 2.8%–12.1% by 2050, impacting future food security (Bernatchez et al., 2017; Costello et al., 2020; Food and Agriculture Organization of the United Nations [FAO], 2018). Consequently, knowledge of the changes in the distribution range and genetic diversity of species and populations is fundamental for the sustainable management of fishery resources (Allendorf et al., 2008; Cheung et al., 2010; Möllmann et al., 2021; Nielsen et al., 2021; Pinsky & Palumbi, 2014).

Species distribution models (SDMs) can be used to predict changes in distributional range into the future, using a variety of tools to predict future climate conditions, such as modeling climate velocity (i.e., determining the rate and direction of climate shifts across the landscape, Pinsky et al., 2013) and rates of isotherm shift (Poloczanska et al., 2013). However, SDMs only rarely consider intra-specific population dynamics such as population genomic variation or divergence, which can limit their potential for informing strategies to protect species' evolutionary dynamics (Lovrenčić et al., 2022; Nielsen et al., 2021). Integrating population genomics approaches with SDM provides additional insights for assessing species' evolutionary potential and characterizing landscape effects on connectivity between populations (Rochat et al., 2021; Wagner & Fortin, 2013). Distributional changes such as range expansion and habitat fragmentation can affect genotypic composition through microevolutionary processes such as migration, genetic drift, and selection. Migration provides new allelic variants that promote genetic diversity, while genetic drift tends to reduce genetic diversity in small populations where alleles can more easily reach fixation. Meanwhile, selection may act in favor of or against locally adapted genotypes. Within the context of changing climates it essential to assess changes in genotypic composition for management issues because it is most beneficial (evolutionarily and economically) to identify and conserve individuals with greater phenotypic plasticity (caused by multiple genetic signals) and to protect animal genetic

resources within the framework of food security (Benestan, 2019; Bernatchez, 2016; Bernatchez et al., 2017).

The power of RAD-seq and associated methods for the detection of both neutral loci and loci that are putatively subject to selection (i.e., outlier loci) has shown great promise in unraveling environment-genome associations in non-model species (Waits & Storfer, 2015). However, some challenges remain, especially in identifying which loci are under selection and determining their ecological relevance (Ahrens et al., 2018; Hoban et al., 2016). In marine species, which can have high dispersal potential, large population sizes, and high levels of gene flow, distinguishing between neutral loci and selected loci may reveal fine-scale evolutionary dynamics (Benestan et al., 2015; Carreras et al., 2017; Lamichhaney et al., 2012). One advantage of using putatively selected loci in genomic population studies is that they may reveal processes that lead to changes in allele frequencies on ecological time scales that neutral loci cannot resolve; that is, loci that are under selection are expected to respond to mechanisms that promote genomic divergence in local populations (Jakobsdóttir et al., 2011; Therkildsen et al., 2013). Furthermore, metrics of genomic variation based on putatively selected genomic regions (rather than neutral markers) may be more effective as indicators of evolutionary potential and as predictors of the spatial distribution of adaptive variants under the effects of climate change, for example, using simulations based on genotype-environment associations (GEAs) (Landguth, Bearlin, et al., 2017; Landguth, Holden, et al., 2017; Meek et al., 2023; Rochat et al., 2021).

The Almaco jack *Seriola rivoliana*, Valenciennes, 1833, is widely distributed circumglobally in warm waters ranging from the Indo-West and Central Pacific to the Eastern Pacific, as well as the Western and Eastern Atlantic and Mediterranean Sea (Castrìota et al., 2002; Eschmeyer & Herald, 1999; Smith-Vaniz, 1984). Throughout its range, *S. rivoliana* forms part of commercial fisheries and in some regions also has high aquaculture potential (Blacio et al., 2003; Quiñones-Arreola et al., 2015; Roo et al., 2014, 2015). However, it is not always targeted directly, leaving knowledge gaps on the extent of exploitation, since for many countries the fishing statistics are reported generically (Comisión Nacional de la Pesca [CONAPESCA], 2021; [FAO], 2021).

In the Pacific Ocean, the FAO has recognized the Eastern-Central Pacific and Pacific Southeast as major fishing areas (FAO, 2015). Furthermore, the Eastern-Central Pacific is considered a highly productive area, with 3000t of catch reported in 2018 alone (FAO, 2021). At the same time, climate change projections in these regions have high uncertainty compared with other regions (see Melo-Merino et al., 2020; Poloczanska et al., 2013), particularly in higher latitudes and extratropical extremes (such as off the coasts of Baja California and Ecuador), where there are also issues related to degradation of coastal ecosystems (FAO, 2018). These high levels of uncertainty make it especially difficult to predict species' responses to climate changes in the Eastern-Central Pacific (Poloczanska et al., 2016).

Seriola rivoliana has a primarily coastal distribution spanning at least 11 countries in the Pacific Ocean. It is therefore essential to define population units for their adequate future management, particularly where resources are shared between countries.

Unfortunately, little is known about the broader genomic structure and diversity of *S. rivoliana*, but using a suite of neutral single-nucleotide polymorphisms (SNPs), Mendoza-Portillo et al. (2020) identified three populations in the Pacific Ocean and found high levels of gene flow between them. These populations did not follow a pattern of isolation by distance, suggesting that environmental variability may be a driver of population divergence and could be responsible for genomic structuring. If this is the case, then analyses based on loci that are under selection (as opposed to neutral loci) and that take into account the influence of environmental variables could better explain the patterns of diversity and genomic structure in this pelagic species.

Given the signals of the population structure of *S. rivoliana* in the Pacific Ocean, which some evidence suggests may be driven by environmental heterogeneity (Mendoza-Portillo et al., 2020), our main objective was to use loci that are putatively under selection (i.e., outlier loci) to test population structure as well as to evaluate the potential impacts of climate change and fishing pressure on the future distribution of genomic diversity and structure of *S. rivoliana* in the Eastern-Central Pacific. Specifically, we aimed to (1) identify the contemporary diversity and structure of *S. rivoliana* populations based on putatively adaptive loci, (2) investigate the potential functions of those loci, (3) evaluate changes in distribution (i.e., range expansions or reductions) using SDMs, testing mitigation, stabilized and high-emissions scenarios of climate change (proposed by Intergovernmental Panel on Climate Change), (4) evaluate how fishing activity influences the connectivity of *S. rivoliana* populations, and (5) evaluate the combined effects of climate change and fishing pressure on genomic diversity and spatial genomic composition using demogenetic simulations. We predict that future climate change scenarios and fishing pressure will impact *S. rivoliana* genomic structure and genotypic composition and lead to a loss of genomic diversity in populations distributed in the Eastern-Central Pacific Ocean, which could have profound effects on fisheries that rely on this resource.

2 | MATERIALS AND METHODS

2.1 | Functional annotation and population genomics

2.1.1 | Sampling and calling of single-nucleotide polymorphisms

Eighty-one individuals distributed throughout the Eastern Pacific Ocean from Mendoza-Portillo et al. (2020) were used for this study (Table S1, Figure 1). Briefly, fin tissue samples of *S. rivoliana* were recorded and DNA was extracted with the salt extraction technique (Sambrook et al., 1989). Genomic libraries were generated based on the 3RADseq method (Bayona-Vásquez et al., 2019) using *Clal*, *MspI*, and *BamHI* restriction enzymes. Paired-end sequencing was performed on an Illumina HiSeq 4000 platform. Raw sequences were demultiplexed and trimmed to 140bp using the *process_radtags*

function in Stacks v2.2 (Catchen et al., 2013). Then, the dDocent v.2.7.3 pipeline (Puritz et al., 2014) was used to call and genotype SNPs using default parameters and a depth of 6X using the *S. rivoliana* genome (GenBank accession number: GCA_002994505) as the reference genome. We applied the following filters using VCFtools v.0.1.14 (Danecek et al., 2011): minimum sequence quality of 30, minimum genotype call rate per locus of 70%, minimum minor allele frequency of 0.03, and exclusion of linked loci ($R^2 > .8$), following the recommendations of Ahrens et al., 2018 and O'Leary et al., 2018. In this study we used the 4938 SNPs identified by Mendoza-Portillo et al., 2020, which represented ~0.00074% of the *S. rivoliana* genome, prior to Hardy-Weinberg filtering to identify loci putatively under selection.

2.1.2 | Outlier loci and gene annotation

We evaluated three independent methods to identify SNPs that were putatively under selection (hereafter, "outlier"). One method was based on principal component analysis (PCA) in the *pcadapt* package for R (Luu et al., 2017), and two were GEA methods: BayeScEnv (De Villemereuil & Gaggiotti, 2015) and redundancy analysis (RDA, Forester et al., 2016, 2018). For *pcadapt*, the value of *K* was set to the number of patches sampled, where patches were defined as buffered sample sites with similar environmental conditions; see Table S1. For the GEA methods, we retained the maximum and minimum values of uncorrelated oceanographic variables (further details are provided below in the section "selection of oceanographic variables"). The BayeScEnv method integrates an F_{ST} -based approach with environmental variables and was executed with 10 pilot runs with 1000 iterations, a thinning interval of 1000 outputted iterations and a burn-in length of 100,000 MCMC, with the convergence of model evaluated with the CODA package in R (Plummer et al., 2006). The RDA method links the genomic structure with environmental factors using a multiple regression model that evaluates the relation between genotypes and environmental predictors (Forester et al., 2018). RDA was carried out in the *vegan* 2.4 package in R (Oksanen, 2017). For all three methods, the *p*-values of selected SNPs were corrected using the false discovery rate (FDR) method (Ahrens et al., 2018; Benjamini & Hochberg, 1995). Three data sets were obtained based on outlier SNPs detected by at least two methods: (1) SNP_{MAX} (SNPs detected by maximum environmental values), (2) SNP_{MIN} (SNPs by minimum environmental values), and (3) SNP_{ALL} (including SNPs detected by both minimum and maximum environmental values).

To evaluate the potential functional roles of outlier loci, the SNPs detected by maximum and minimum values were blasted in National Center for Biotechnology Information (NCBI) database for functional annotation. We considered 1000 base pairs upstream and downstream of each identified outlier SNP. The analyses were run in Blast2Go (OmixBox 1.4, Conesa et al., 2005), which conducts BLAST similarity searches and maps Gene Ontology (GO). It was performed with default parameters, incorporated "Actinopterygii" filter, and a predetermined *e*-value cutoff of 10^{-5} .

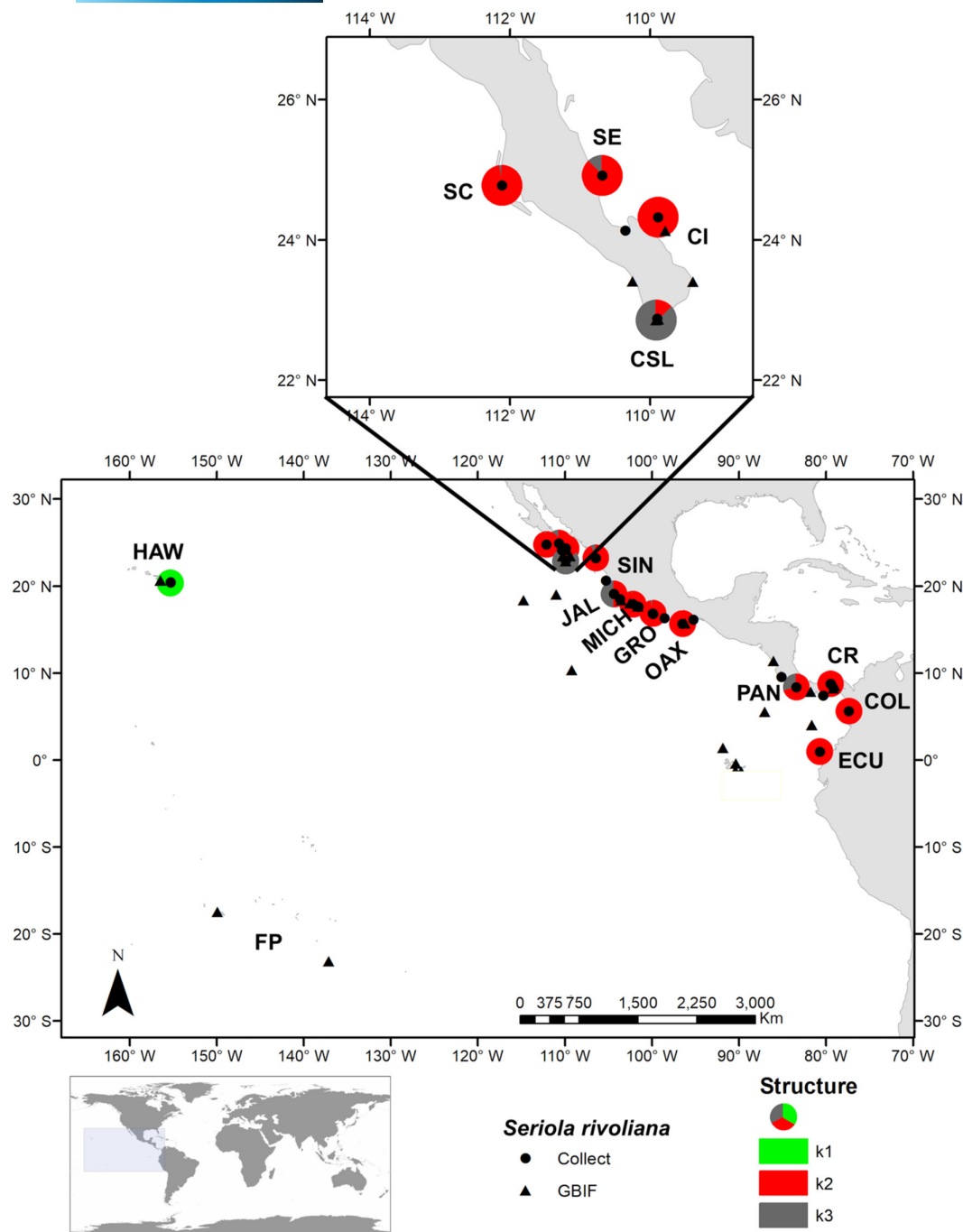


FIGURE 1 Localities of *Seriola rivoliana* obtained from Global Biodiversity Information Facility, GBIF (▲) and collected data (●). The pie graphs indicate the ancestry coefficient obtained by STRUCTURE (estimated with 155 outlier loci: SNP_{ALL}), which each color indicating the genomic population (k), to which belongs. Acronyms: S0an Carlos (SC), San Evaristo (SE), Cerralvo Island (CI), Cabo San Lucas (CSL), Sinaloa (SIN), Jalisco (JAL), Michoacán (MICH), Guerrero (GRO), Oaxaca (OAX), Panama (PAN), Costa Rica (CR), Colombia (COL), Ecuador (ECU), Hawaii (HAW), and French Polynesia (FP). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/gcb.16732)]

2.1.3 | Estimating genomic diversity and population structure

As sample sizes differed among the 23 collection sites, we defined “patches” by applying a buffer in ArcMap 10.3 (Environmental Systems Research Institute (ESRI), 2013) to group sites with similar environmental conditions within a radius of 100 km, which was then evaluated

by PCA following Stein et al. (2015). This resulted in 14 patches. We then used the three SNP data sets of outlier loci for downstream analyses. Individual ancestries were analyzed using two methods: STRUCTURE 2.3.3 (Pritchard et al., 2000, 2009) and discriminant analysis of principal components (DAPCs) using the *Adegenet* package in R (Jombart, 2008). STRUCTURE was performed in a hierarchical manner to identify putative populations within groups inferred from previous

runs following Pritchard et al. (2009) and Vähä et al. (2007). Each run of STRUCTURE was carried out in STRAUTo (Chhatre & Emerson, 2017) under the admixture model and correlated allele frequencies. We used a burn-in of 10,000 and 100,000 MCMC steps and k values between 1 and 14, with 10 replicate runs for each value of k . The optimal k value was determined by the change of mean log likelihoods of the data, $\ln P(D)$ (Pritchard et al., 2000), and rate of change in $\ln P(D)$, Δk (Evanno et al., 2005), with results compiled and visualized using CLUMPAK (Kopelman et al., 2015). For DAPC, the cross-validation function was applied to identify the number of principal components to retain.

We also calculated genomic diversity measures—rarefied allelic richness (A_r), allelic frequency, expected (H_e) and observed (H_o) heterozygosity and inbreeding coefficient (G_{is})—in GENODIVE v.3.04 (Meirmans, 2020) to identify differences among the genomic populations. The genomic differentiation was evaluated using Hedrick's pairwise standardized measure (G''_{ST} , Meirmans & Hedrick, 2011) and the F_{ST} estimator, which were calculated with 1000 permutations in GENODIVE v.3.04. p -values were adjusted using the FDR method. To make the estimates of genomic differences comparable between contemporary patches and future climatic and fishing pressure scenarios (see below), Mantel tests (Mantel, 1967) were carried out based on G''_{ST} comparisons.

2.2 | Spatial distribution models and future projections

2.2.1 | Occurrence data

Occurrence data for *S. rivoliana* comprised 23 sampling locations from this study and 31 occurrences sites from the Global Biodiversity

Information Facility (Global Biodiversity Information Facility (GBIF), Gbif.org, 2020) database (Table S1 and Figure 1). All records were spatially filtered, retaining only one point per pixel at the standard marine spatial resolution of 5 arcmin (~9.2 km, Assis et al., 2018).

2.2.2 | Selection of oceanographic variables

Maximum, minimum, and mean values of 13 oceanographic variables were considered under two approaches: physicochemical and biological association (Table 1). The oceanic information was taken from several online databases between 2013 and 2018 (the years of sampling in this study), including monthly chlorophyll and diurnal and nocturnal sea surface temperature from Aqua-MODIS (<http://oceancolor.gsfc.nasa.gov/>) and physicochemical variables from Bio-ORACLE (Assis et al., 2018). The information from each variable was extracted with ARCMAP 10.3 using the database of occurrence data. All oceanographic variables were scaled, then we excluded highly correlated variables ($\rho \geq 0.8$, Nogueira et al., 2017) based on all pairwise Spearman's correlations, calculated using the *stats* package in R (R Core Team, 2018). Multicollinearity among variables was evaluated through the variance inflation factor (VIF) with "*vi*" function of the *usdm* package in R (Naimi, 2015); variables with $VIF > 10$ were excluded (Akinwande et al., 2015). This yielded a total of eight maximum and seven minimum uncorrelated variables used for the GEA method for detecting outlier loci, and a total of seven non-correlated mean variables for constructing the spatial distribution models (Table 1 and Table S2). For future climate scenarios, the environmental variables available (temperature, salinity, and current velocity [CV]) projected for the years 2050 and 2100 were obtained from the Bio-ORACLE repository (Assis et al., 2018), under three Representative Concentration Pathways (RCPs): the mitigation scenario RCP2.6 (low

TABLE 1 Thirteen oceanographic variables used for the identification of outlier loci through genetic–environmental association method and as predictors for the spatial distribution models.

Oceanographic variable	Description	Unit	Source	VIF (max)	VIF (min)	VIF (mean)
DA*	Diffuse attenuation	m^{-1}	Bio-Oracle	2.594	2.382	3.790
CHL	Chlorophyll	$mg \cdot m^{-3}$	NASA-AquaModis	—	—	—
CV	Current velocity	m^{-1}	Bio-Oracle	1.706	1.953	1.741
O2	Dissolved oxygen	$mol \cdot m^{-3}$	Bio-Oracle	2.995	2.726	—
N	Nitrate	$mol \cdot m^{-3}$	Bio-Oracle	77.641	—	4.175
P	Phosphate	$mol \cdot m^{-3}$	Bio-Oracle	2.719	4.68.4	12.117
SA*	Salinity	PSS	Bio-Oracle	1.929	—	—
DST*	Diurnal sea surface temperature	°C	NASA-AquaModis	2.911	6.806	9.050
NST	Night sea surface temperature	°C	NASA-AquaModis	—	—	—
PHY	Phytoplankton	$\mu mol \cdot m^{-3}$	Bio-Oracle	—	—	—
PP	Primary productivity	$g \cdot m^{-3} \cdot day^{-1}$	Bio-Oracle	1.959	1.976	2.237
BAT	Bathymetry	m	Bio-Oracle	5.914	3.033	7.818
PH	pH	—	Bio-Oracle	—	—	4.724

Note: For each variable, the variance inflation factor by sub-data set is shown: maximum, minimum, and mean values. The "—" indicates the available variables for future projections.

In bold are values that indicate the variables excluded by multicollinearity ($VIF > 10$).

greenhouse gas concentration levels); the stabilized scenario RCP4.5 and the high-emission scenario RCP8.5 (emissions increases over time, Assis et al., 2018).

2.2.3 | Calibration and validation of models

Spatial distribution models based on occurrence (GBIF and collecting records) and environmental (uncorrelated oceanic variables) data were used to infer habitat suitability under the contemporary and projected future climate scenarios. We used several approaches for the SDMs: generalized linear model, generalized additive model, classification tree analysis, artificial neural networks (ANNs), surface range envelope, flexible discriminant analysis, multiple adaptive regression splines, generalized boosting model (GBM), random forest (RF), and two versions of maximum entropy (MAXENT). All of these models were run in the *biomod2* 3.4.6 package in R (Thuiller, 2003; Thuiller et al., 2020). For all algorithms, two data sets of 1000 random points pseudo-absence points each were generated (PA1 and PA2) within the proposed range dispersal range following Barbet-Massin et al. (2012). The data were split into calibration (70% of the data) and validation (30%) data sets, and each algorithm was run for 10 replicates, resulting in a total of 220 models (11 algorithms $\times$ 2 sets of random points $\times$ 10 replicates). Values equal to or greater than 0.8 of the true skill statistic (TSS) and the area under the receiver operating characteristic curve (ROC) were chosen to evaluate model performance. Predicted habitat suitability for the retained models were concatenated to produce a weighted mean probability SDM of contemporary and future scenarios, which were projected using ArcMap 10.3.

2.2.4 | Changes between contemporary and future distribution models

The spatial changes of suitable habitat between contemporary and future climate scenarios were calculated with the *minus* raster function in ArcMap 10.3 using the probability maps obtained from SDM (Table S3). Habitat expansion, contraction, and unchanged areas were estimated and plotted at 1 pixel = 84.64 km² (5 arcmin resolution, Assis et al., 2018).

2.3 | Fishing pressure

As climate change affects the range of species distribution and this could affect fishing patterns, fishing pressure was included as a spatial resistance matrix that estimates the connectivity of *S. rivoli-ana*. As fishing information for *S. rivoli-ana* is largely missing both for Mexico and other study areas, information about the fishing activity, as well as other boat data and tools to monitor global fishing patterns were obtained from Global Fishing Watch (<https://globalfishingwatch.org>) and used as a proxy of fishing pressure. Fishing effort data (fishing hours) were downloaded from 2012 to 2017, and we

generated a raster layer that included the average fishing effort following Kroodsma et al. (2018) at a 5 arcmin pixel resolution. The average fishing effort in hours was calculated with the *statistics* toolbox in ArcMap 10.3, and the results were homogenized into classes of resistance data. The classes were based on the quantile distribution on a scale from 1 (low pressure or free movement) to 10 (high pressure or no movement). This raster layer was used to perform connectivity analysis as the fishing pressure scenario.

2.4 | Connectivity

The habitat suitability scores obtained from the SDM of contemporary and future scenarios (Table S3) were inverted ($\text{SDM} \times \text{inverted} = 1 - x$) in SDMtoolbox 2.4 (Brown, 2014) in ArcMap 10.3 to convert the areas of high suitability to areas of low dispersal cost, resulting in seven resistance surfaces or “friction maps” (Wang et al., 2008). Friction maps quantify the strength of dispersal barriers at any given point in space, that is, the difficulty of dispersal, assigning each pixel a value between 1 (low resistance to dispersal) and 10 (high resistance). A total of eight friction maps (seven SDM and fishing pressure, Table S3) were considered as input data into CIRCUITSCAPE v4.0 (McRae & Shah, 2011). The pairwise resistance measures among patches were calculated under eight-neighbor parameter in all resistance surfaces in CIRCUITSCAPE. Finally, all connectivity scenarios were plotted as connectivity maps in ArcMap 10.3.

2.5 | Changes in structure and genomic diversity under different scenarios

To project changes in the structure and genomic diversity of *S. rivoli-ana* populations under different connectivity scenarios, demogenetic simulations were carried out in CDMetaPOP (Landguth, Bearlin, et al., 2017). Briefly, CDMetaPop simulates changes in genotype frequencies over time taking into account demographic information (i.e., growth, movement, reproduction, mortality) of individuals located in patches (geographic sites) in heterogeneous landscapes (resistance surfaces or environmental gradients), in addition to incorporating microevolutionary processes such as gene flow, drift, mutation, and bottlenecks from empirical or parametrized genetic information (for more details, see Landguth & Cushman, 2010; Landguth, Bearlin, et al., 2017).

2.5.1 | Demogenetic parameters

To reduce problems due to unequal sample sizes among the patches and their effect on genetic drift, the simulations were run under two sample sizes—500 (M1–M4 models, Table S4) and 10 individuals (M5–M8 models, Table S4) with 50% males and 50% females per patch with heterosexual reproduction and a generation time of 4 years (age at maturity *S. rivoli-ana*, Farmer et al., 2016). We used dispersal distance

as a measure of gene flow, using the maximum dispersal distance known in the genus *Seriola* was incorporated (40% of the available space, ~3400 km reported for *Seriola lalandi*; Gillanders et al., 2001). Due to the lack of tracking information for mating distances in *S. rivoliana*, we used several different percentages of the maximum dispersal distance: 1% (M1, M5), 10% (M2, M6), 40% (M3, M7), and 70% (M4, M8), which were simulated with a random mixing function. The values of dispersal resistance were made comparable with each other by standardizing them to a mean value of zero and unit variance in R (Becker et al., 1988). To estimate the additive pressures of fishing and climate change scenarios, the resistance layer for fishing pressure was overlapped with each one of the climate change scenarios using the *plus* function in ArcMap 10.3. The allelic frequencies of putative adaptive loci obtained in this study (SNPs_{ALL} data set) were used as input data. Generational allelic frequencies were calculated every 4 years (approximate generational time in *Seriola* genus, Roo et al., 2014; Zupa et al., 2017) from 2018 to the year 2100 under eight scenarios of heterogeneous landscapes: (1) isolation by distance (no resistances, i.e., the null model), (2) fishing pressure (FP), (3) 2050-RCP2.6 plus fishing pressure effect (+FP), (4) 2050 RCP4.5 + FP, (5) 2050 RCP8.5 + FP, (6) 2100 RCP2.6 + FP, (7) 2100 RCP4.5 + FP, and (8) 2100 RCP8.5 + FP which were run with modified sample size and mating distances (M1–M8, for complete parameters used, see Table S4). Each scenario under each model was run for 10 replicates, following Landguth and Cushman (2010).

Expected heterozygosity (H_e) was used to evaluate the differences between the different simulated models, mainly between the sample sizes, with ANOVA in the *stats* package for R. The genetic results of the simulation were analyzed to calculate genomic structure under DAPC approach and pairwise differentiation among patches through Hedrick's standardized G''_{ST} estimations in GENODIVE v.3.04. Then, a Mantel test was executed to compare the changes in genomic differentiation (G''_{ST}) from different scenarios respect to empirical data in GENODIVE v.3.04.

3 | RESULTS

3.1 | Functional annotation and population genomics

3.1.1 | Sampling and calling of single-nucleotide polymorphisms

A total of 300 outlier SNPs were identified, all of them through *pca-dapt* (Figure S5). For minimum values of environmental variables, BayeScEnv identified 84 SNPs while RDA detected 59 SNPs; for maximum environmental values 94 and 51 SNPs by BayeScEnv and RDA were detected, respectively. Only seven SNPs were identified by three methods considering minimum and maximum environmental values (Figure S5). From all outliers, those identified by at least two methods were selected, with 138 SNPs for SNP_{MAX}, 136 SNPs for SNP_{MIN}, and 155 SNPs for SNP_{ALL}.

3.1.2 | Outlier loci and gene annotation

Seventy-eight of 155 candidate SNPs (from SNP_{ALL} data set) had significant BLAST matches (Table S6a). Most of the sequences were related to Greater amberjack (*Seriola dumerili*), Tanaka's snailfish (*Liparis tanakae*) and Yellowtail amberjack (*S. lalandi dorsalis*, Table S6b). Fifty-five SNPs were assigned under different protein annotations (Table S6), with most assigned to biological processes (BP, 54.19%) related to cellular and metabolic processes (Table S6c); 30.17% were related to molecular functions, mainly to binding and catalytic activity. The cellular component accounted for 15.64% of gene ontology, which was mostly represented by cellular anatomical entities (Table S6c).

3.1.3 | Genomic diversity and population structure

The clustering methods identified three genomic populations ($k=3$, Figure S7) for all SNP data sets. The DAPC accounted for more than 99% of the genomic variation in the first two discriminant functions (DFs) in all data sets (Figure S7a). The first group or putative population contained all individuals sampled in Cabo San Lucas; the second group contained all individuals from Eastern Pacific from Cerralvo Island to Ecuador, including individuals from San Carlos; and the third group contained the individuals from Hawaii. The hierarchical structure analyses revealed two groups in the first run that separated Hawaii from the Eastern Pacific patches, and the second run, which included only Eastern Pacific individuals revealed $k=2$, separating out individuals from Cabo San Lucas. Thus, there was full concordance between the STRUCTURE and DAPC analyses (Figure S7b).

For genomic diversity, the rarefied allelic richness (A_r) was similar among all populations ($A_r=1.201: 1.256$, Table 2). No significant differences in expected (H_e) or observed (H_o) heterozygosity were observed between populations in any of the three SNP data sets (SNP_{MAX}, SNP_{MIN} and SNP_{ALL}, Table S8a). In contrast, the Hawaii population showed significant differences by presenting low levels of inbreeding coefficient ($G_{is}=0.005: 0.029$) with respect to the Eastern Pacific populations (Cabo San Lucas: $G_{is}=0.186: 0.235$, $p<.02$; Eastern Pacific: $G_{is}=0.117: 0.205$, $p<.003$, Table S8b).

Since the results of clustering and genomic diversity did not detect differences among the three SNP data sets (Table S8), only the SNP_{ALL} data set (155 SNPs) was used for further analyses estimating genomic differentiation and allelic frequency in the simulation analysis. Estimates of F_{ST} and G''_{ST} supported the three genomic populations detected through DAPC and STRUCTURE; the first population contained individuals from Hawaii, with strong genomic differentiation from the Eastern Pacific patches ($F_{ST}>0.76$, $G''_{ST}>0.9$, $p<.05$, Table S9a); the second differentiated population corresponded to the Cabo San Lucas patch ($F_{ST}=0.24-0.35$, $G''_{ST}=0.23-0.58$, $p<.05$ with respect to the Eastern Pacific locations), and the third population was consisted of the remaining patches in the Eastern Pacific, which did not show significant values of F_{ST} . Although estimates of G''_{ST} suggested genomic differentiation between San Evaristo with Jalisco, Oaxaca, Costa

TABLE 2 Measures of genomic diversity in populations for each dataset of outliers single-nucleotide polymorphisms (SNPs) detected (138 SNP_{MAX}, 136 SNP_{MIN}, and 155 SNP_{ALL}).

Population	Patch	SNP _{MAX} AR	SNP _{MAX} H_e (CI 95%)	SNP _{MAX} H_o (CI 95%)	SNP _{MAX} G_{is} (CI 95%)	SNP _{MIN} AR
1 (CSL)	CSL	1.201	0.058 (0.04–0.078)	0.045 (0.031–0.060)	0.235 (0.096–0.363)	1.202
2 (EPA)	SC, CI, SIN, JAL, COL, MICH, GRO, OAX, PAN, CR, COL, ECU	1.205	0.058 (0.041–0.078)	0.046 (0.031–0.064)	0.205 (0.094–0.323)	1.221
3 (HAW)	HAW	1.239	0.103 (0.072–0.133)	0.101 (0.068–0.135)	0.017 (–0.135–0.167)	1.256

Abbreviations: AR, Allelic richness (rarefied), H_e , expected, H_o , observed heterozygosity and inbreeding coefficient (G_{is}).

Rica, Colombia, and Ecuador ($G''_{ST}=0.17-0.22$, $p<.05$, Table S9a); and Guerrero with Michoacán, Oaxaca, Colombia, and Ecuador ($G''_{ST}=0.14-0.20$, $p<.05$), this was not supported by the F_{ST} analysis.

3.2 | Spatial distribution models and future projections

3.2.1 | Selection of oceanographic variables

After testing for correlations between oceanographic variables, night sea surface temperature and chlorophyll a concentration were removed from all data sets ($\rho \geq 0.8$). The remaining maximum variables were as follows: diffuse attenuation (DA), CV, dissolved oxygen (O_2), phosphate (P), salinity (SA), diurnal sea surface temperature (DST), primary productivity (PP), and bathymetry (BAT). The minimum variables were: DA, CV, O_2 , DST, PP, BAT, and phytoplankton (PHY). The mean data set variables, used to construct the spatial distribution models, were as follows: DA, CV, DST, PP, BAT, nitrate (N), and pH (Table S2 and Table 1).

3.2.2 | Spatial distribution model calibration and validation

The spatial distribution model algorithms that had the best values of the ROC and TSS indicators (≥ 0.8 , Table S10) were the ANNs, GBM, generalized linear model (GLM), and RF, which were used to build the contemporary and future projection models (Table S3). The results of these algorithms were assembled to generate a final consensus model to project the probability of a contemporary suitable area of *S. rivoliana* (Figure 2) and each future climate scenario (Table 3, Table S3).

3.2.3 | Changes in future distribution models

Of the environmental variables, DA contributed most strongly to the SDM prediction for the contemporary scenario (46.94%), followed by salinity (13.37%). For the future projections, salinity had the strongest contribution with 68.12% (Table S11). The SDM for the contemporary scenario revealed that the best suitable distribution range for *S. rivoliana* is between 20°N and 10°S, with a disruption in suitable habitat

near the coast of central Mexico and in the Central Pacific between 130° and 150°W (Figure 2).

Changes in habitat suitability in future scenarios with respect to the SDM for the contemporary scenario are shown in Table 3 and Table S12. The SDMs for all scenarios projected range contractions mainly from the south of the Gulf of California to Hawaii, although some northward expansion was also shown, especially in scenarios for 2050 (Table S12a,c,e). Central America was the most stable zone, although there was a slight expansion that intensified under the RCP8.5 scenario (Table S12e).

As expected, comparing the three RCP scenarios, the RCP2.6 scenario had the highest percentages of areas of stable habitat suitability (i.e., areas that were suitable both in the current and future scenarios) for both 2050 (38.88%) and 2100 (45.54%) as well as the lowest percentages of range contraction with 33.89% for 2050 and 37.37% for 2100 (Table S12a,b and Table 3). In contrast, the RCP8.5 scenario showed the greatest increase in suitable areas for both 2050 (38.73%) and 2100 (24.51%) at the same time as having the greatest area of contraction of suitable habitat (44.18%) by the year 2100 (Table S12e,f and Table 3). It is noteworthy that under all climate change scenarios in 2050, the probability of habitat suitability was greater than 0.90, comparable with contemporary habitat suitability probability (0.91), but by the year 2100, the probability was reduced to 0.85 (RCP8.5, Table 3).

3.3 | Fishing pressure

Estimated fishing effort from 2012 to 2017 ranged between 0.5 h (several regions in Pacific) and 116 h (some areas between Hawaii and French Polynesia) per day. The sites with most fishing activity in the Eastern Pacific were the southern Central Pacific with up to ~48 h per day, southwest Mexico (near the coast of Chiapas) with ~32 h and the Baja California Peninsula with average of ~16 h, while Hawaii had fishing effort of ~29 h (Figure S3b).

3.4 | Connectivity

The CIRCUITSCAPE analyses (Figure 3) provide estimates of the regions with high connectivity and a higher probability of movement of individuals (and therefore more likely gene flow). For the contemporary

SNP _{MIN} H_e (CI 95%)	SNP _{MIN} H_o (CI 95%)	SNP _{MIN} G_{is} (CI 95%)	SNP _{ALL} AR	SNP _{ALL} H_e (CI 95%)	SNP _{ALL} H_o (CI 95%)	SNP _{ALL} G_{is} (CI 95%)
0.064 (0.044–0.087)	0.052 (0.036–0.071)	0.186 (0.074–0.299)	1.244	0.073 (0.053–0.093)	0.056 (0.041–0.073)	0.224 (0.114–0.329)
0.067 (0.047–0.088)	0.055 (0.038–0.072)	0.177 (0.098–0.256)	1.252	0.075 (0.055–0.096)	0.060 (0.044–0.078)	0.196 (0.111–0.286)
0.107 (0.075–0.138)	0.107 (0.072–0.141)	0.005 (–0.130–0.142)	1.251	0.104 (0.075–0.132)	0.101 (0.07–0.132)	0.029 (–0.109–0.161)

scenario (Figure 3a), as well as for the scenarios modeled for 2050 (climate change plus fishing pressure, Figure 3c–e), there was high connectivity between localities near the Baja California peninsula westward nearly to Hawaii. However, the connectivity is fragmented to the north of the Baja California Peninsula, as well as Central America, with higher intensity under the most severe climate change scenario in combination with fishing pressure (RCP8.5 + FP, Figure 3e).

Populations along the coastline from the Gulf of California to Ecuador maintained strong connectivity under both the contemporary and mitigation scenarios when including fishing pressure (RCP2.6 + FP, Figure 3a,c, respectively). However, this connectivity is lost under RCP4.5 and RCP8.5 scenarios in combination with fishing pressure (Figure 3d,e). In addition, in the scenario where only fishing pressure was simulated, the interruption of connectivity in different places along the coastline is more evident, mainly from southern Mexico to Ecuador (Figure 3b). Otherwise, it is important to note that no connectivity was observed between the central Pacific locations, between Hawaii and the Polynesian Islands, or from the Central Pacific to the Eastern Pacific.

3.5 | Changes in structure and genomic diversity versus different scenarios

The models (M1–M4) with sample sizes of 500 individuals were not significantly different from each other ($p > .05$) (Table S13), although model M3 (capacity of 40% of the distance for mating) was the most homogeneous (Table S13b). To determine the influence of sample size (10 vs. 500 individuals) on the genomic structuring estimated by DAPC models, we compared between M3 (Figures 4 and 5) and M7 (Table S14) because they both used the same simulated distance parameters (~1340 km, which is equivalent to 40% maximum reported distance 3400 km), which is consistent with that reported for other *Seriola* species (Gillanders et al., 2001; Tone et al., 2022). Under the null scenario, isolation by distance, the two models showed the same panmictic pattern for all patches, including the most genetically distinct group in Hawaii (Figure 4b and Table S14). For the fishing pressure only scenario, M3 showed four genomic groups: two well differentiated (Hawaii, Cabo San Lucas), and two interconnected in Eastern Pacific coast (Figure 4c). Meanwhile, M7 showed additional clusters along the Eastern Pacific coast (Table S14). For all simulations with climate change in addition to fishing pressure scenarios, M7 consistently showed Hawaii as an

isolated genomic cluster and had a larger number of genomic clusters in the Eastern Pacific compared with M3 (Table S14).

Simulations with M3 showed a clustering pattern similar to fishing pressure, except the mitigation scenario in 2050 (2050 RCP2.6 + FP) where Hawaii does not appear as an isolated cluster (Figure 5a). Within the Eastern Pacific, the simulations showed some differences, mainly in the Cabo San Lucas patch, which was always shown in the lower left quadrant of the DAPC DF, except in the simulations for 2050 (RCP4.5 + FP), where it did not differ from the rest of the Eastern Pacific and for the 2100 simulations (RCP8.5 + FP), where it occurred in the upper left quadrant of the DAPC (Figure 5). Table S9 shows the G''_{ST} values for the different patches represented in the DAPC analysis.

Three patterns of expected heterozygosity (H_e) were found with the M3 simulations (Table S15): H_e similar to that reported by empirical data in the San Evaristo, Sinaloa, Cabo San Lucas, Colombia, and Costa Rica patches; increased H_e compared with the contemporary scenario in Panama and Ecuador, and loss of H_e in San Carlos, Cerralvo Island, Oaxaca, Michoacán, Jalisco, and Hawaii, with the last two patches presenting the highest levels of contemporary genomic diversity ($H_e = 0.087$, $H_e = 0.203$, respectively). Finally, Mantel tests did not show significant correlations between contemporary (empirical) genomic differentiations versus different climate change and fishing pressure scenarios (r Mantel statistic range = 0.07–0.29, $p > .03$, Table S9b).

4 | DISCUSSION

Understanding the evolutionary trajectories of marine species under future scenarios that include climate change and other anthropogenic pressures such as fishing, is vital for supporting contemporary decision-making that can help safeguard the long-term resilience of marine species (Möllmann et al., 2021). In this study, we evaluate how future climatic change and fishing pressure may influence the future diversity, genomic structure, and connectivity of *S. rivoliana*, an important fishery species in the Central and Eastern Pacific Ocean. Using loci that are putatively subject to natural selection, we show that contemporary patterns of genomic structure fall into three genomic populations—two in the Eastern (Cabo San Lucas, and Eastern Pacific) and one in the Central Pacific (Hawaii). The outlier loci identified were related mostly to the biological and metabolic processes that may be related to temperature and salinity. Future projections suggest a loss of suitable habitat and potential range contractions for most scenarios. Importantly, the changes in population connectivity under

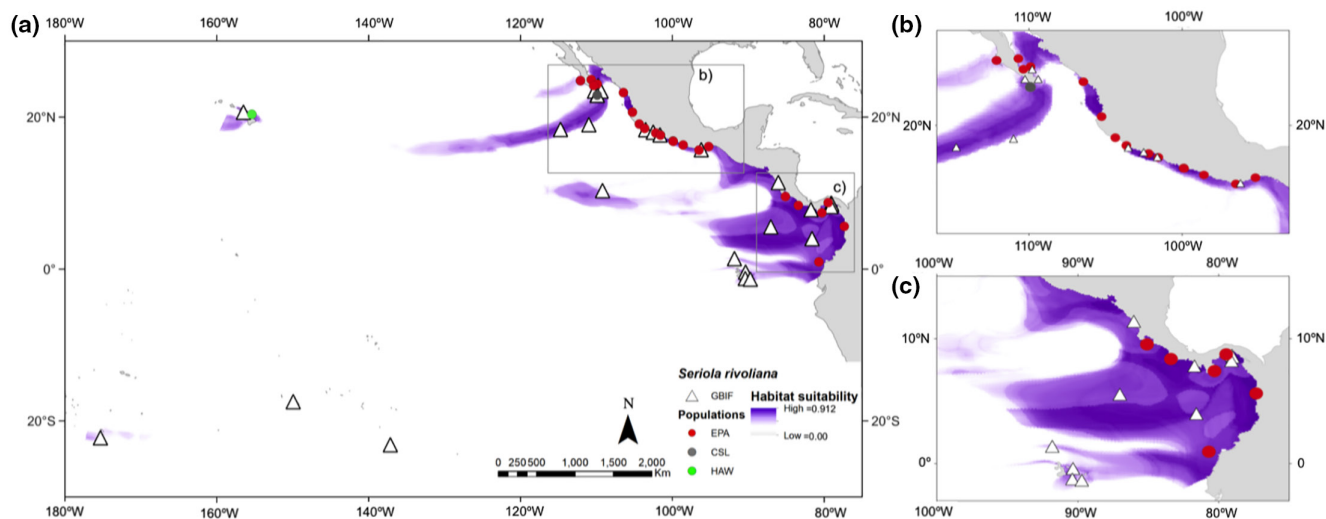


FIGURE 2 Contemporary probability of habitat suitability of *Seriola rivoliana* obtained from spatial distribution models (a). Records of *Seriola rivoliana* obtained from Global Biodiversity Information Facility, GBIF (▲) and collected data (●). Colors represent the genetic population. Zoom in on areas where GBIF and collection site symbols are overlapped for Mexico (b) and Central America (c). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/gcb.16732)]

TABLE 3 Changes of expansion, contraction, or no changes in habitat suitability (area and percentage) respect contemporary conditions in different climate change scenarios for the years 2050 and 2100.

Scenario	Max probability of habitat suitability	Range expansion (respect all area available)		Range contraction (respect all area available)		Range of no changes	
		Area (km ²)	%	Area (km ²)	%	Area (km ²)	%
Contemporary	0.91	—	—	—	—	—	—
2050 RCP 2.6	0.91	1,161,991.75	27.23	1,446,034.18	33.89	1,659,066.00	38.88
2050 RCP 4.5	0.90	1,316,923.98	30.86	1,620,332.94	37.97	1,329,835.01	31.16
2050 RCP 8.5	0.91	1,652,610.49	38.73	1,504,133.77	35.25	1,110,347.67	26.02
2100 RCP 2.6	0.91	729,472.60	17.1	1,594,510.90	37.37	1,943,108.43	45.54
2100 RCP 4.5	0.87	742,383.62	17.4	1,775,265.18	41.60	1,749,443.13	41.00
2100 RCP 8.5	0.85	1,045,792.58	24.51	1,885,008.84	44.18	1,336,290.51	31.32
Contemporary occupancy area (km ²)		4,267,091.93 (100%)					

different climatic, fishing pressure, and combined scenarios were highly correlated with the simulated genomic structure. As such, we find that fishing pressure is an important variable that will likely affect the patterns of population genomic structuring (Table S14) and genotypic composition of *S. rivoliana* under climate change.

4.1 | Population genomic structure and functional annotation of outlier SNPs

Blast2GO analysis revealed 55 SNPs located in genes or regions of known genes (SNP_{ALL}, Table S6a). Most of the genes identified are involved in cellular processes, metabolism, and catalytic activity (Table S6c) reported for other fish species, mainly members of the genus *Seriola* (Table S6b). These results suggest that these SNPs may play important roles in local adaptation since a large proportion of the annotated loci are related to membrane structure that could have

importance in processes such as osmoregulation, which likely interact with environmental factors such as salinity (Kalujnaia et al., 2007). Importantly, although only 155 outlier SNPs were found and only about a third of these could be associated with a known function (a small number relative to genome size), a single region may be associated with the regulation of a variety of signal transduction pathways, which are involved in the immunity, growth, and development (Dalongeville et al., 2018). Despite limitations in the representativeness of putatively selected loci, this approach has proven to be useful to reveal local adaptation. For example, in *Penaeus monodon* and *Leuciscus waleckii*, SNPs associated with salt stress were detected in transcriptional regulators (Si et al., 2022; Wang et al., 2021).

Although uncertainties remain in the detection of SNPs associated with local adaptation, outlier detection methods, and their downstream analyses are attracting growing interest in the basic science and environmental management communities. While these genomic approaches are not easy to implement for many wild populations

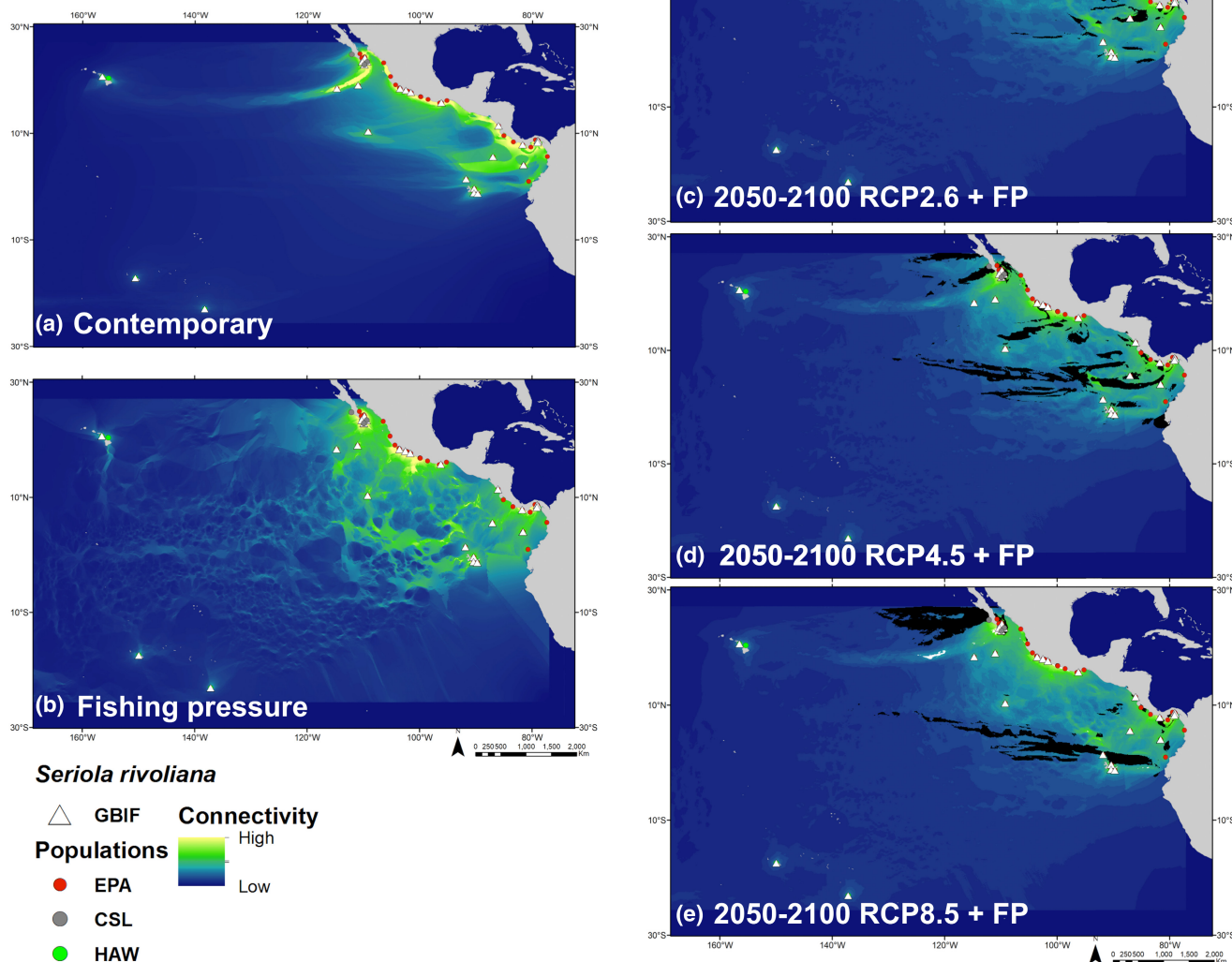


FIGURE 3 Connectivity maps (left panel) for contemporary (a), fishing pressure (b), under RCP2.6 + FP (c), under RCP4.5 + FP (d) and RCP8.5 + FP (e). The yellow gradient shows the intensity of connectivity among different patches in the Pacific Ocean while black areas represent as loss of connectivity between 2050 and 2100. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/gcb.16732)]

(particularly those lacking annotated genomes), they offer the opportunity to refine efforts for the management and conservation of species and to identify the populations with the greatest risk of decline due to climate and other anthropogenic changes (Hoffmann et al., 2015; Meek et al., 2023; Nielsen et al., 2023). Further experimental research and in situ studies are required to verify the importance of these gene regions for local adaptation.

In this study, the results of adaptive genomic differentiation between the sampled patches of *S. rivoliana* showed higher values of F_{ST} than those previously reported based solely on neutral markers (Mendoza-Portillo et al., 2020). There are still several uncertainties around the detection of loci that are under selection that may influence disparate patterns between neutral and outlier loci. First, false positives in the detection of outlier loci can be minimized by using

multiple detection methods and incorporating environmental variation (Ahrens et al., 2018; François et al., 2016; Hoban et al., 2016; Nielsen et al., 2020). In our study, we reduce false positives following the recommendations of François et al. (2016), by incorporating three different outlier detection methods, spanning minimum and maximum values of environmental variables (Ahrens et al., 2018). Another issue is related to environmental clines, where atypical loci associated with environmental gradients show a structured pattern along similar environmental values (Frichot et al., 2015; Hoban et al., 2016). Under that scenario, the Hawaiian and Cabo San Lucas patches would be expected to be genetically close as both areas show similar salinity values (Table S16); this was not the case, rejecting the hypothesis that genomic structure follows a salinity gradient. Given that salinity was similar between patches,

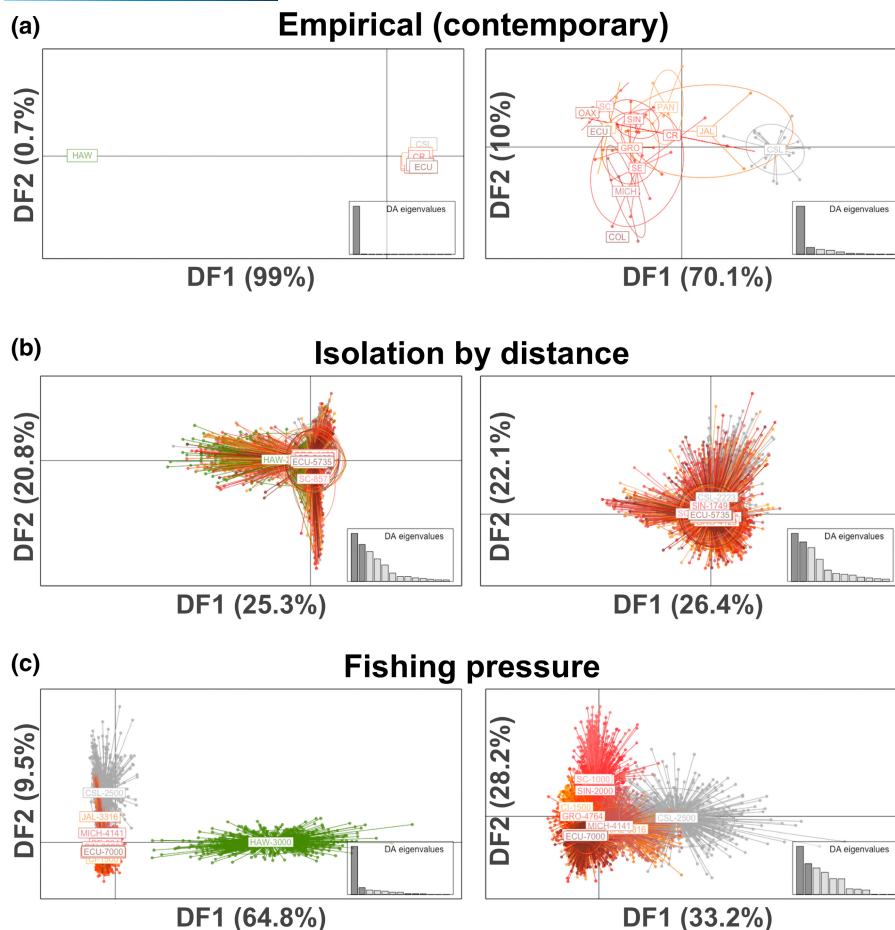


FIGURE 4 Scatterplots showing the genomic clusters identified by DAPC of empirical data (a), and scenarios simulated under model M3 (500 individuals per patch) in CDMetaPOP. Left: plots considering all patches, right: plots excluding Hawaii to visualize better the structure inside Eastern Pacific. Scenarios: Isolation by distance (b), Fishing pressure (c). All simulations were run with 155 outlier loci (SNP_{ALL}). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/gcb.16732)]

the influence of other variables such as temperature and chlorophyll also drives environmental heterogeneity, including between the Warm Temperate Northeast Pacific and Hawaii marine ecoregions (Tzeng, 2022). It is therefore reasonable to assume that the genomic structure reported for *S. rivoliana* coincides with the patterns of heterogeneity of these environmental variables. Another difference in genomic structure assessed with outlier loci compared with neutral loci is that the closest genomic relationship was between San Carlos and the Eastern Pacific stock, rather than between San Carlos and Cabo San Lucas, as was reported with neutral markers. This pattern could be explained by chaotic genetic patchiness (CGP), which is common in marine species, particularly in the region of the Pacific under study (e.g., Sardines and Anchovies, Lecomte et al., 2004). The GCP is likely associated with climatic fluctuations (Jiménez-Quiroz et al., 2019; Lecomte et al., 2004; Rodríguez-Romero et al., 2011) and selection, either after population establishment due to variation in environmental conditions or prior to establishment, which produces variability in cohorts across space and time (Eldon et al., 2016; Hogan et al., 2010). It is important to emphasize that there are other factors such as migration, drift, and life history traits that promote genetic variation that could generate CGP (Eldon et al., 2016). These mechanisms are

not mutually exclusive, and more research is required to unravel this pattern of genomic differentiation in the vicinity of the Baja California peninsula.

4.2 | Spatial distribution models and future projections

Shifts in the geographic distribution of *S. rivoliana* in 2050 and 2100 showed redistribution to higher latitudes, which is consistent with results described for other marine species, including both pelagic (Cheung et al., 2009; García Molinos et al., 2016; Jones & Cheung, 2015; Poloczanska et al., 2016) and demersal fishes (Fredston-Hermann et al., 2020; Perry, 2005). By 2050, the areas with the best habitat suitability for *S. rivoliana* are found near the Baja California peninsula. Although some studies show that areas near the tropics will be the most affected in future scenarios due to increasing temperatures (García Molinos et al., 2016; Jones & Cheung, 2015; Mora, Frazier, et al., 2013), the potential distribution of *S. rivoliana* in the Central American zone remained stable under all climate change scenarios (Tables S3 and S12). This is likely linked to future salinity projection for this region for 2050 and 2100, which are similar to

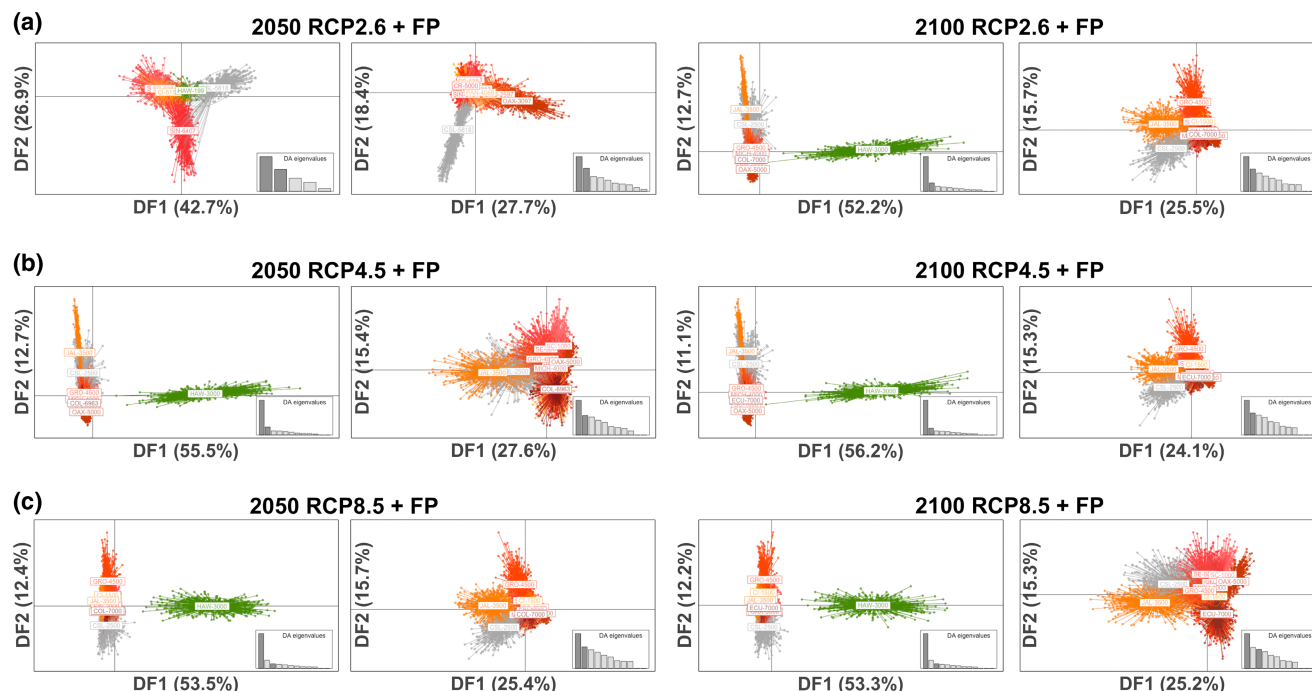


FIGURE 5 Scatterplots showing the genetic clusters identified by DAPC in climate change and fishing pressure scenarios, under RCP2.6 + FP (a), RCP4.5 + FP (b), and RCP8.5 + FP (c) for 2050 (left panel) and 2100 (right panel) under M3 model (500 individual per patch) in CDMetaPOP. Left plots consider all patches, whereas those on the right exclude Hawaii and, thus, better visualize the structure in the Eastern Pacific. All simulations were run with 155 outlier loci (SNP_{ALL}). [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/gcb.16732)]

contemporary values (between 31.1 and 32 ppm; Table S16, Assis et al., 2018; Tyberghein et al., 2012). The projected changes in habitat suitability reported in this study are mainly due to salinity and temperature changes, which according to Melo-Merino et al. (2020) are commonly used in marine SDM. This is partly because these two variables are easy to measure and widely available in international databases but also because they have strong biological relevance for marine species (e.g., Nielsen et al., 2020). For *S. rivoliana*, salinity and temperature best predict the suitability of future habitat, accounting for ~68.12% and 18.11%, respectively (Table S11). However, as is the case for many marine species, evolutionary and physiological plasticity will probably differ between juvenile and adult individuals and as such, future climate change may affect different life history stages in different ways. For example, Reinoso et al. (2019) reported that survival of *S. rivoliana* larvae is affected when the salinity is very low or very high because larvae are forced to allocate more energy to osmoregulation, at the expense of growth and development. Temperature is an important determinant of the incubation times of *S. rivoliana* eggs, as well as the phenotypes associated with its embryonic development. Viader-Guerrero et al. (2021) show that the viable temperature for this species ranges between 22°C and 24°C, but subtle changes in temperature for this species affect the hatching success, survival, growth, and development of the larvae, warmer temperatures accelerated hatching, conversely, lower temperatures (<22°C) reduce survival, with a hatching rate low to zero (Viader-Guerrero et al., 2021). In this study, we found different loci that could be associated with cellular as well as regulatory and metabolic

processes (Table S6) that may be important for adapting to changing temperature and salinity conditions. Further studies aimed at better understanding the effects of environmental variation (and therefore potential future scenarios) on different ontological stages of this commercially important species are crucial.

4.3 | Connectivity

As expected, the loss of connectivity was more evident under the RCP4.5 and RCP8.5 scenarios, particularly when combined with fishing pressure (Figure 3). The most affected areas were east of Central America and northeast of the Baja California peninsula. The connectivity represented by the suitability of the future habitat is a good proxy to understand the possible routes of movement under future climate scenarios. Some authors have stated that the inverse of suitability maps is a poor basis for inferring dispersal movements because mobile animals can more easily cross-habitats of lower quality, which allows the identification of corridors (Keeley et al., 2017). However, for aquatic species, it is difficult to track movement, particularly during early life-history stages, since they are additionally affected by physical factors such as oceanic currents. As such, although the connectivity models implemented in this study have some limitations (e.g., they are unable to account for the complex interplay of biotic and abiotic variation), in the absence of precise knowledge, they nonetheless present valid biological estimates for predicting optimal geographic areas and levels of connectivity between them (Dickson et al., 2019).

4.4 | Implications for changes in structure and genomic diversity under different scenarios

Changes in connectivity under different climate scenarios and fishing pressures are highly correlated with the simulated genomic structure in *S. rivoliana* populations. Furthermore, we show that geographic distance is not a strong predictor of genomic differentiation, but that connectivity is limited by fishing pressure and climatic changes. This drives stronger genomic structure in the interior of the Eastern Pacific compared with the contemporary genomic structure (Figures 4 and 5). Across almost all modeled scenarios, Hawaii was consistently genetically distinct and isolated from populations in the Eastern Pacific. Therefore, it should be considered an independent population unit or stock, and within the context of aquaculture, should not be mixed with individuals from other regions.

Although fishing pressure strongly determined patterns of future genomic structure in our models, we found that there are subtle changes when combined with climate scenarios. For example, the Cabo San Lucas patch appears as a genomic cluster on the horizontal axis of the DAPC plots, but its position along the vertical axis changed (Figure 5). This could be related to the level of genotypic variability of individuals that are distributed in the vectors as a function of space, according to the scenario modeled. We observed that the homozygosity of some loci changed with the simulation scenarios (Table S17). For example, we found a reduction in the homozygosity of two loci associated with the genomic regions (SNP_1151 associated with the BS transposon DNA polymerase, with transferase activity, and the SNP_1211 locus associated with the polypyrimidine tract-binding protein involved in the regulation of RNA splicing, Table S6), showing a response of genotypic changes to the climatic scenarios tested. Our results are also consistent with the lower heterozygosity reported for future climate change scenarios (Pauls et al., 2013) and with the report by Pinsky and Palumbi (2014) that overfishing strongly reduces heterozygosity. Overfishing has been shown to potentially reduce the climate resilience of exploited fish species. For example, in the sparid *Chrysoblephus laticeps*, individuals with better aerobic performance were preferentially removed from fished populations (Duncan et al., 2019), and as such fishing itself may act as a selective pressure, with the selective removal of certain phenotypes generating changes in allele frequencies (Jakobsdóttir et al., 2011; Therkildsen et al., 2013).

Furthermore, the combination of overfishing and climate change could spatially homogenize genetic diversity, which could have negative effects on locally adapted populations, as genetic drift and climatic stressors could act to reduce adaptive potential. Allele surfing—the spread of rare alleles in wider areas where the alleles can reach very high frequencies—can occur as a consequence of drift, which can severely alter genetic variability at neutral and adaptive loci, causing low genomic diversity (Excoffier & Ray, 2008). Conversely, migration could increase the abundance of adaptive genotypes by maintaining or even increasing connectivity between populations (Pauls et al., 2013). Here, we discovered that most simulations combining fishing pressure and climate change change not

only the level of genomic structure but also its genotypic composition and that such changes may leave some areas with increased genomic vulnerability (Nielsen et al., 2021).

Although the inclusion of fishing pressure provided a more nuanced perspective of the genomic future of *S. rivoliana*, including reducing connectivity, we faced some limitations in terms of data availability. For this study, we relied primarily on global vessel data because there are generally no records of specific catches for *S. rivoliana* at the local or international level, and when there were (e.g., in Mexico), they are reported with the generic name “scale fishing,” which includes more than 200 species of fishes (Diario Oficial de la Federación (DOF), 2017). This makes it difficult to incorporate changes in fishing pressure into models of future scenarios and is an important consideration for the collection and dissemination of fishing activity data (Möllmann et al., 2021).

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

The genotypes from the 155 outlier loci data and input files for CDMetaPop are available in Dryad: <https://doi.org/10.5061/dryad.zgmsbccd1>.

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